

AKLILU BEYERO

Software Engineer | Machine Learning and AI Trainer

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PROFILE

Full-stack developer and AI trainer with a degree in Computer Engineering from Addis Ababa University. I've been working in software for over five years, building web applications for startups and clients, and training AI models on real software tasks across multiple programming languages. Also experienced in cybersecurity research, software security, and vulnerability analysis, with a background in analyzing vulnerable code, reproducing security flaws, developing proof-of-concept exploits, implementing secure patches, and validating fixes using fuzzing-based testing techniques. Comfortable working with C/C++, Python, Linux, debugging tools, and secure software development practices. Strong understanding of memory safety vulnerabilities, input validation, and secure coding principles.

SKILLS

Programming & Frameworks: Python, C++, JavaScript, C, TypeScript, ReactJS, Next.js, NodeJS, React Native, Flask, TensorFlow, PyTorch (applied usage), HTML, CSS, Tailwind CSS

Machine Learning & Deep Learning: Deep Learning, Convolutional Neural Networks (CNNs), Transfer Learning, Model Training & Evaluation, Hyperparameter Tuning, Dataset Preprocessing, Computer Vision Applications

Backend & Databases: RESTful APIs, MongoDB, MySQL, PostgreSQL, Sequelize

Tools & Platforms: Git, GitHub, Supabase, Slack, Windows, Linux (basic)

Embedded & Engineering Tools: MATLAB, Proteus, Arduino, IAR Embedded Workbench

PYTHON / ML ENGINEERING EXPERIENCE

AI Training & Evaluation — Revelo

- Performed data labeling and annotation for machine learning dataset development
- Evaluated AI model outputs for correctness, quality, and edge-case behavior
- Designed and refined prompts to test model robustness and failure cases
- Contributed to improving dataset quality through consistent evaluation standards
- Applied structured reasoning to identify model weaknesses and inconsistencies

Cybersecurity Experience

- Designed intentionally vulnerable code repositories for security testing and AI evaluation
- Injected OSS-Fuzz-style vulnerabilities into C/C++ and Python applications for controlled testing environments
- Used fuzzing techniques to reproduce crashes and identify memory corruption, input validation, and parsing vulnerabilities
- Developed secure patches to eliminate discovered vulnerabilities while preserving application functionality
- Verified fixes through regression testing and repeated fuzzing campaigns
- Analyzed crash reports, root causes, and exploitability
- Worked with AddressSanitizer (ASan), UndefinedBehaviorSanitizer (UBSan), GDB, and Linux debugging tools
- Followed secure coding practices aligned with modern vulnerability research workflows

PROFESSIONAL EXPERIENCE

Full Stack Engineer — Amplitude Ventures AS

May 2024 – Jun 2026

Full-time · Norway · Remote

- Built and maintained full-stack features using React.js and Angular for production web applications
- Worked across software infrastructure, contributing to scalable and maintainable system architecture
- Collaborated with a distributed, remote team to design, develop, and ship web development projects
- Applied a broad range of skills spanning frontend, backend, and infrastructure development

Full-Stack Developer

Oct 2024 – Jan 2025

- Developed and maintained full-stack web applications using modern frameworks

- Built responsive user interfaces and integrated backend APIs
- Implemented server-side logic and database interactions
- Participated in debugging, testing, and performance improvements
- Collaborated in team-based development workflows using Git

Electrical Engineering Intern

Oct 2024 – Jan 2025

- Supported planning and execution of electrical infrastructure projects
- Assisted in system installation, testing, and maintenance activities
- Conducted site inspections and technical troubleshooting
- Prepared documentation and field reports
- Ensured safety and compliance standards in field operations

PROJECTS

Real-Time Plant Disease Detection System (Deep Learning)

- Developed CNN-based image classification model for plant disease detection
- Trained and evaluated deep learning models using image datasets
- Performed preprocessing, augmentation, and model tuning to improve accuracy
- Integrated trained model into a mobile application for real-time inference

E-Commerce Web Application

- Built a full-stack e-commerce platform supporting users and vendors
- Implemented authentication, product management, cart, and order workflows
- Developed backend APIs and integrated database systems
- Focused on usability, performance, and system reliability

Hospital Management System

- Designed and developed a web-based system for hospital operations
- Implemented patient records, appointments, and staff management modules
- Built backend services and database structure for scalable data handling

Soil Moisture Detection System

- Built embedded system using LPC2138 microcontroller
- Implemented soil moisture monitoring for irrigation efficiency
- Designed hardware-software integration for real-world sensor usage

EDUCATION

BSc in Electrical and Computer Engineering (Computer Engineering)

2021 – 2025

Addis Ababa University

Relevant Coursework: Data Structures & Algorithms, Database Systems, Computer Networks, Machine Learning, Computer Architecture, Computer Security

LANGUAGES

English (Fluent)

INTERESTS

Machine learning systems, deep learning, AI evaluation, software engineering, building scalable applications, continuous learning, and solving real-world technical problems.